CLASS 9 · CBSE · GANITA MANJARI PART 1 · SHORT NOTES

Chapter 8: Predicting What Comes Next? Exploring Sequences and Progressions

Topic-wise short notes with solved examples

A sequence is an ordered list of numbers. This chapter shows two ways of writing its rule, and studies two special sequences: the arithmetic progression, which grows by adding, and the geometric progression, which grows by multiplying.

1 Sequences and their terms

- A **sequence** is an ordered list of numbers. Each number is a **term**.
- A sequence is **finite** if it stops (6, 12, 24, 48, 96) and **infinite** if it goes on for ever. Three dots show that it continues.
- The terms are written $t_1, t_2, t_3, \ldots$ The subscript is the position: $t_4 = 7$ means the fourth term is 7. The position n is always a natural number, but the term itself can be any number.

Sequence	Terms	Pattern
Natural numbers	$1, 2, 3, 4, 5, \ldots$	each term is 1 more than the last
Odd numbers	$1, 3, 5, 7, 9, \ldots$	the difference is always 2
Triangular numbers	$1, 3, 6, 10, 15, \ldots$	$1,\ 1+2,\ 1+2+3, \ldots$
Square numbers	$1, 4, 9, 16, 25, \ldots$	$1,\ 1+3,\ 1+3+5, \ldots$

- The nth square number is the sum of the first n odd numbers. The nth triangular number is the sum of the first n natural numbers.
- Some sequences, such as the prime numbers $2, 3, 5, 7, 11, \ldots$, have no simple rule.

2 Explicit rule

- An **explicit rule** gives the nth term directly from its position n. Any term can be found without knowing the earlier ones.
- To check whether a number is a term, set the rule equal to the number and solve for n. It is a term only if n is a natural number.

Example 2.1 For the sequence $t_n = 4n + 1$, find the first three terms and the 25th term. Are 213 and 150 terms of the sequence?

SOLUTION

$t_1 = 5$, $t_2 = 9$, $t_3 = 13$ and $t_{25} = 101$

$4n + 1 = 213$ gives $n = 53$. $4n + 1 = 150$ gives $n = 37.25$.

5, 9, 13; $t_{25} = 101$; 213 is the 53rd term; 150 is not a term

3 Recursive rule

- A **recursive rule** gives the first term (or first few terms) and a way to get each term from the terms before it.
- To use it, the earlier terms must be worked out first.

Example 3.1 Find the first four terms of the sequence $u_1 = 2$, $u_n = 3u_{n-1} - 1$ for $n \geq 2$.

SOLUTION

$u_2 = 3(2) - 1 = 5$, $u_3 = 3(5) - 1 = 14$, $u_4 = 3(14) - 1 = 41$

2, 5, 14, 41

- The **Virahānka–Fibonacci sequence** uses the two previous terms: $V_1 = 1$, $V_2 = 2$ and $V_n = V_{n-1} + V_{n-2}$ for $n \geq 3$. It gives $1, 2, 3, 5, 8, 13, 21, 34, 55, 89, \ldots$

REMEMBER

Virahānka wrote down this sequence in the 7th century CE while studying the metres of Prakrit poetry. Gopāla and Hemachandra studied it further, well before Fibonacci (c. 1200 CE).

4 Arithmetic progressions

- An **arithmetic progression (AP)** is a sequence in which each term is obtained by **adding** a fixed number to the term before it. The fixed number is the **common difference** d.
- General form: $a,\ a + d,\ a + 2d,\ a + 3d,\ \ldots$ where a is the first term.

$$t_n = a + (n - 1)d$$

- Recursive rule: $t_1 = a$ and $t_n = t_{n-1} + d$ for $n \geq 2$.
- d can be negative: $11, 7, 3, -1, \ldots$ is an AP with $d = -4$.
- Plotting the points (n, t_n) of an AP gives points on a **straight line**. An AP is a linear pattern.

Example 4.1 Find the nth term and the 20th term of the AP $7, 11, 15, 19, \ldots$

SOLUTION

$a = 7$ and $d = 4$, so $t_n = 7 + (n - 1) \times 4 = 4n + 3$.

$t_n = 4n + 3$; $t_{20} = 83$

Example 4.2 Which term of the AP $20, 17, 14, \ldots$ is -40? Is 0 a term of this AP?

SOLUTION

$t_n = 20 + (n-1)(-3) = 23 - 3n$

$23 - 3n = -40$ gives $n = 21$. $23 - 3n = 0$ gives $n = \frac{23}{3}$, which is not a natural number.

-40 is the 21st term; 0 is not a term

Example 4.3 The 4th term of an AP is 17 and the 9th term is 37. Find the AP and its 15th term.

SOLUTION

$a + 3d = 17$ and $a + 8d = 37$. Subtracting: $5d = 20$, so $d = 4$ and $a = 5$.

$t_{15} = 5 + 14 \times 4$

The AP is $5, 9, 13, \ldots$ and $t_{15} = 61$

Example 4.4 How many three-digit numbers are divisible by 6?

SOLUTION

They form the AP $102, 108, \ldots, 996$ with $d = 6$.

$996 = 102 + (n-1) \times 6$, so $n - 1 = 149$.

150 numbers

5 Sum of the first n natural numbers

- Write the sum forwards and backwards and add:

$$S = 1 + 2 + \ldots + n \qquad S = n + (n-1) + \ldots + 1$$

- Each of the n pairs adds up to $n + 1$, so $2S = n(n+1)$.

$$S_n = 1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}$$

- Āryabhaṭa stated this result in the Āryabhaṭīya (499 CE): the sum is the average of the first and last terms multiplied by the number of terms.
- The nth **triangular number** is $\frac{n(n+1)}{2}$.
- A sum that does not start from 1 is a difference of two such sums.

Example 5.1 Find $1 + 2 + 3 + \ldots + 40$ and the 12th triangular number.

SOLUTION

$S_{40} = \frac{40 \times 41}{2}$ and $\frac{12 \times 13}{2}$

820 and 78

Example 5.2 Find $31 + 32 + 33 + \ldots + 70$.

SOLUTION

The sum is $S_{70} - S_{30} = \dfrac{70 \times 71}{2} - \dfrac{30 \times 31}{2} = 2485 - 465$

2020

6 Geometric progressions

- A **geometric progression (GP)** is a sequence in which each term is obtained by **multiplying** the term before it by a fixed number. The fixed number is the **common ratio** r.
- General form: $a,\ ar,\ ar^2,\ ar^3,\ \ldots$

$$t_n = ar^{n-1}$$

- Recursive rule: $t_1 = a$ and $t_n = r \times t_{n-1}$ for $n \geq 2$.
- To test for a GP, check that the ratio of each term to the one before it is the same.
- r can be a fraction or negative: $1, -1, 1, -1, \ldots$ is a GP with $r = -1$.
- Plotting the points (n, t_n) of a GP does **not** give a straight line. The points rise (or fall) faster and faster.

Example 6.1 Is $8, 12, 18, 27, \ldots$ a GP? If so, find its nth term.

SOLUTION

$\dfrac{12}{8} = \dfrac{3}{2}, \dfrac{18}{12} = \dfrac{3}{2}, \dfrac{27}{18} = \dfrac{3}{2}$. The ratio is constant.

Yes, with $a = 8$ and $r = \dfrac{3}{2}$: $t_n = 8\left(\dfrac{3}{2}\right)^{n-1}$

Example 6.2 Find the 7th term of the GP $2, 6, 18, \ldots$ Which term of the GP $5, 10, 20, \ldots$ is 1280?

SOLUTION

$t_7 = 2 \times 3^6 = 2 \times 729$

$5 \times 2^{n-1} = 1280$, so $2^{n-1} = 256 = 2^8$ and $n = 9$.

1458; the 9th term

Example 6.3 A culture starts with 50 bacteria and the number doubles every hour. How many are there after 6 hours and after n hours?

SOLUTION

After each hour the number is multiplied by 2, so after n hours it is 50×2^n.

3200 after 6 hours; 50×2^n after n hours

Example 6.4 A ball is dropped from a height of 50 m and each bounce reaches 80% of the previous height. How high does it rise after the 4th bounce?

SOLUTION

The heights form a GP with first term $50 \times 0.8 = 40$ and $r = 0.8$: 40, 32, 25.6, 20.48.

20.48 m

7 Fractals and geometric progressions

- A **fractal** is a pattern that repeats itself at smaller and smaller scales.
- **Sierpiński triangle**: start with a triangle (stage 0). At each stage, join the mid-points of the sides of every shaded triangle and remove the middle one.

Stage n	0	1	2	3	n
Number of shaded triangles	1	3	9	27	3^n
Shaded area (stage 0 has area 1)	1	$\frac{3}{4}$	$\frac{9}{16}$	$\frac{27}{64}$	$\left(\frac{3}{4}\right)^n$

- The number of triangles is a GP with ratio 3 and grows quickly. The shaded area is a GP with ratio $\frac{3}{4}$ and gets closer and closer to 0.

8 AP and GP compared

	AP	GP
How the next term is found	add d	multiply by r
What stays constant	difference $t_n - t_{n-1}$	ratio $\frac{t_n}{t_{n-1}}$
nth term	$a + (n-1)d$	ar^{n-1}
Graph of (n, t_n)	points on a straight line	points not on a straight line

9 Common mistakes

- Writing $t_n = a + nd$. The first term has no d added, so it is $a + (n-1)d$.
- Writing the nth term of a GP as ar^n. The power is $n - 1$.
- Accepting a value of n that is not a natural number. Then the number is not a term.
- Taking the common difference as (earlier term − later term). It is $t_2 - t_1$.
- Calling a sequence a GP after checking only one ratio. Check each pair of consecutive terms.